

Revealed Preferences

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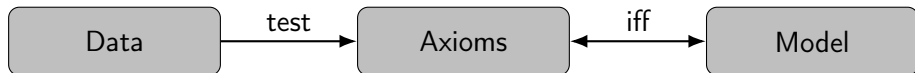
Goal

- Review the revealed-preference machinery from first-year PhD micro.
- Emphasize the choice-theoretic perspective: data, model, axioms, representation.

Why?

Most behavioral models add latent objects: attention, temptation, reference points, beliefs, search costs, mistakes. Revealed preference tells us what those latent objects imply for observable choice.

The Basic Logic



- The model has unobservable objects: utility, preferences, marginal rates of substitution.
- A representation theorem gives the exact observable restrictions of the model.
- Best case: the axioms are necessary and sufficient.

Why This In Behavioral Economics?

- Behavioral economics often relaxes the standard model by adding psychological mechanisms.
- These mechanisms are usually not directly observed.
- The point is not whether a model has a plausible story.
- Instead: what choice patterns does it permit, forbid, or distinguish?

To keep in mind

Without a representation theorem, a richer behavioral model may be descriptively attractive but empirically empty: it may explain almost anything after the fact.

Data: Choice Correspondence

Definition

Let X be a finite set of alternatives. A choice correspondence is a map

$$C : 2^X \setminus \{\emptyset\} \rightarrow 2^X \setminus \{\emptyset\}$$

such that $C(A) \subseteq A$ for every nonempty $A \subseteq X$.

- A is the choice set.
- $C(A)$ is the set of chosen alternatives from A .
- Multi-valued choice allows indifference or ties.

Model: Utility Maximization

Definition

A utility function $u : X \rightarrow \mathbb{R}$ rationalizes C if for every nonempty $A \subseteq X$,

$$C(A) = \arg \max_{x \in A} u(x) = \{x \in A : u(x) \geq u(y) \forall y \in A\}.$$

- The utility numbers are latent.
- The choice correspondence is observable, at least in the idealized data set.
- The problem is to characterize when such a u exists.

Axioms: α , β , and WARP

Axiom α

If $x \in B \subseteq A$ and $x \in C(A)$, then $x \in C(B)$.

Axiom β

If $x, y \in C(A)$, $A \subseteq B$, and $y \in C(B)$, then $x \in C(B)$.

WARP

If $x, y \in A \cap B$, $x \in C(A)$ and $y \in C(B)$, then $x \in C(B)$.

Representation Theorem

Theorem

A choice correspondence C on a finite set X has a utility representation if and only if it satisfies axioms α and β .

- Necessity: utility maximization implies the axioms.
- Sufficiency: the axioms let us construct a rational preference relation and then a utility representation.
- Therefore, α and β are the empirical content of utility maximization in this setting.

Preferences as an Intermediate Object

Definition

A binary relation $\succeq$ on X is a rational preference relation if it is complete and transitive.

Definition

$\succeq$ represents C if

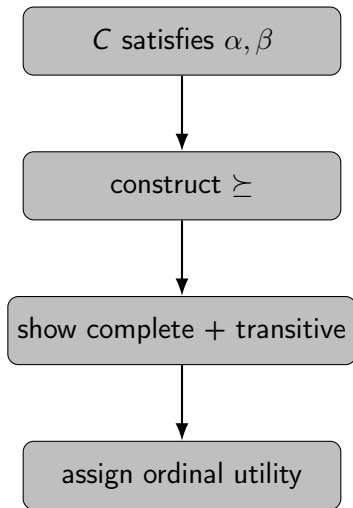
$$C(A) = \{x \in A : x \succeq y \ \forall y \in A\}$$

for every nonempty $A \subseteq X$.

- A utility function represents $\succeq$ when

$$x \succeq y \iff u(x) \geq u(y).$$

Proof Strategy for Sufficiency



Ordinality and Non-Uniqueness

Theorem

Let $u : X \rightarrow \mathbb{R}$ represent C . A function $v : X \rightarrow \mathbb{R}$ also represents C if and only if there exists a strictly increasing transformation T such that

$$v(x) = T(u(x)) \quad \text{for all } x \in X.$$

- Choice identifies rankings, not utility differences.
- $u(x) = 2, u(y) = 3, u(z) = 1$ and
 $v(x) = 0, v(y) = 10000, v(z) = -5000$ can encode the same behavior.
- Cardinal utility requires additional structure beyond ordinal choice.

Three Idealizations in the Basic Data Set

The first theorem assumes:

1. X is finite.
2. Choices are observed from every nonempty subset of X .
3. Choice is a correspondence, so ties are observable.

Each assumption is convenient for the theorem. Each becomes restrictive when we move to actual experiments, market data, or behavioral models.

Infinite Domains: Preference vs Utility

- Choices can be represented by preferences under α and β without finiteness.
- Utility representation is harder on infinite domains.
- Need additional topological assumptions, usually continuity.

Definition

A preference relation $\succeq$ on a metric space X is continuous if whenever $x \succ y$, sufficiently small perturbations x' of x and y' of y still satisfy $x' \succ y'$.

Debreu's Theorem

Theorem (Debreu)

Let X be a separable metric space and let $\succeq$ be a complete preference relation on X . If $\succeq$ is continuous, then it can be represented by a continuous utility function.

- Continuity rules out lexicographic preferences.
- In finite data, continuity cannot be directly violated.
- This matters because most consumer-choice models use continuous commodity spaces.

Existence of Maximal Choices

Weierstrass logic

If A is compact and u is continuous, then $\arg \max_{x \in A} u(x)$ is nonempty.

- In finite choice, existence is automatic.
- In infinite choice, existence depends on compactness/closedness and continuity.
- Consumer theory often needs compact budget sets or additional restrictions to guarantee demand exists.

Partial Data: Why α and β Are Not Enough

Suppose the only observed choices are

$$C(\{x, y\}) = \{x\}, \quad C(\{y, z\}) = \{y\}, \quad C(\{x, z\}) = \{z\}.$$

- No observed pair of nested choice sets generates a violation of α or β .
- But the data imply the cycle $x \succ y \succ z \succ x$.
- No utility representation exists.

The right tool for partial data is revealed preference, not α and β stated only on observed nested sets.

Direct Revealed Preference

Definition

x is directly revealed preferred to y , written xR^Dy , if for some observed choice set A ,

$$y \in A \quad \text{and} \quad x \in C(A).$$

Definition

x is revealed preferred to y , written xRy , if there is a chain

$$xR^Dw_1R^Dw_2R^D\cdots R^Dw_nR^Dy.$$

That is, R is the transitive closure of R^D .

Strict Revealed Preference

Definition

x is strictly revealed preferred to y , written xSy , if for some observed choice set A ,

$$y \in A, \quad y \notin C(A), \quad x \in C(A).$$

- Direct revealed preference records weak comparisons.
- Strict revealed preference records actual rejection when another option is chosen.
- Strictness is essential because indifference is otherwise hard to identify from choice correspondences.

Definition

A choice correspondence satisfies the Generalized Axiom of Revealed Preference if it is never the case that

$$xRy \quad \text{and} \quad ySx.$$

Equivalently,

$$xRy \Rightarrow \neg(ySx).$$

- GARP forbids a revealed-preference cycle with at least one strict link.
- It is the correct consistency condition for incomplete collections of choice sets.

Representation with Partial Choice Data

Theorem

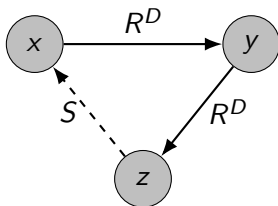
A choice correspondence C defined on an arbitrary collection of nonempty subsets of X satisfies GARP if and only if it has a preference representation.

Corollary

If X is finite, then C satisfies GARP if and only if it has a utility representation.

This is the key revealed-preference result behind consumer-choice applications: we can test rationalizability without observing every possible choice set.

Graph Interpretation



- Acyclicity is not quite enough: weak cycles may reflect indifference.
- GARP fails only when a revealed cycle contains a strict reversal.

The Consumer Choice Environment

- Commodity space: $X = \mathbb{R}_+^n$.
- Prices: $p \in \mathbb{R}_{++}^n$.
- Wealth: $w \in \mathbb{R}_+$.

Definition

The budget set is

$$B(p, w) = \{x \in \mathbb{R}_+^n : px \leq w\}.$$

In consumer theory, the observed choice sets are not arbitrary subsets of $\mathbb{R}_+^n$; they are budget sets generated by prices and wealth.

Demand Correspondence

Definition

Given a preference relation $\succeq$, the demand correspondence is

$$x(p, w) = \{x \in B(p, w) : x \succeq y \ \forall y \in B(p, w)\}.$$

- Demand is homogeneous of degree zero:

$$x(\alpha p, \alpha w) = x(p, w) \quad \text{for all } \alpha > 0.$$

- This follows because $B(\alpha p, \alpha w) = B(p, w)$.

Monotonicity, Local Non-Satiation, Walras' Law

Definition

$\succeq$ is locally non-satiated if for every x and every $\varepsilon > 0$, there exists $y \in B(x, \varepsilon)$ such that $y \succ x$.

Proposition

If demand is represented by locally non-satiated preferences, then it satisfies Walras' Law:

$$px = w \quad \text{for every } x \in x(p, w).$$

If a chosen bundle left money unspent, local non-satiation would allow an arbitrarily nearby improvement within the budget set.

WARP for Demand

Definition

A demand correspondence satisfies WARP if for any $(p, w), (p', w')$,

$$x \in x(p, w), x' \in x(p', w'), p'x \leq w', px' \leq w \Rightarrow x' \in x(p, w).$$

This is the same WARP logic in budget language:

$$x, x' \in B(p, w) \cap B(p', w').$$

If x was chosen when x' was feasible, then choosing x' elsewhere cannot strictly reject x .

Compensated Law of Demand

Definition

Given (p, w) and $x \in x(p, w)$, a compensated price change is (p', w') such that

$$w' = p'x.$$

Definition

Demand satisfies the compensated law of demand if, whenever $x' \in x(p', w')$,

$$(p' - p)(x' - x) \leq 0.$$

If $x' \notin x(p, w)$, then the inequality is strict.

WARP and the Compensated Law

Proposition

Suppose demand satisfies Walras' Law. Then demand satisfies WARP if and only if it satisfies the compensated law of demand.

$$\begin{aligned}(p' - p)(x' - x) &= p'x' - p'x - px' + px \\ &= w' - w' - px' + w \\ &= w - px'.\end{aligned}$$

The compensated law is WARP expressed as a price-change restriction.

Useful Preference Restrictions

Property	Behavioral/economic content
Monotonicity	More of every good is better.
Local non-satiation	No chosen bundle is locally best if nearby improvements are feasible.
Convexity	Mixtures are weakly preferred to extremes.
Strict convexity	Interior mixtures are strictly preferred to extremes.
Homotheticity	Indifference curves expand radially; income expansion paths are simple.
Quasilinearity	One good absorbs income effects; useful for welfare analysis.

Finite Price-Quantity Data

Consumer data often have the form

$$D = \{(p^i, x^i) : i = 1, \dots, N\},$$

where $p^i \in \mathbb{R}_{++}^n$ and $x^i \in \mathbb{R}_+^n$.

- We observe prices and chosen bundles.
- Wealth is usually not observed separately.
- But observed expenditure $p^i x^i$ tells us which bundles would have been affordable at observation i .

Revealed Preference from Budget Data

Definition

x^i is directly revealed preferred to x^j , written $x^i P^D x^j$, if

$$p^i x^i \geq p^i x^j.$$

Definition

x^i is strictly directly revealed preferred to x^j if

$$p^i x^i > p^i x^j.$$

Interpretation: at prices p^i , bundle x^j was affordable when x^i was chosen.

GARP for Consumer Data

Definition

Budget data satisfy GARP if whenever

$$x^i R x^j,$$

it is not the case that

$$p^j x^j > p^j x^i.$$

Equivalently: no sequence of revealed-preference comparisons returns to a bundle that is strictly cheaper than the bundle actually chosen at that final observation.

Afriat's Theorem

Theorem (Afriat)

Let $D = \{(p^i, x^i) : i = 1, \dots, N\}$ be finite price-quantity data. The following are equivalent:

1. D is rationalizable by a locally non-satiated rational preference relation admitting a utility representation.
2. D satisfies GARP.
3. There exist $u^i \in \mathbb{R}$ and $\lambda^i > 0$ such that

$$u^j \leq u^i + \lambda^i p^i (x^j - x^i) \quad \text{for all } i, j.$$

4. D is rationalizable by a continuous, concave, piecewise-linear, strictly monotone utility function.

The Constructed Afriat Utility Function

Assume the Afriat inequalities hold:

$$u^j \leq u^i + \lambda^i p^i (x^j - x^i) \quad \forall i, j.$$

Define

$$u(x) = \min_{i=1, \dots, N} \{ u^i + \lambda^i p^i (x - x^i) \}.$$

- Each term is affine and strictly increasing.
- The minimum of affine functions is concave and piecewise linear.
- The inequalities ensure that $u(x^i) = u^i$ at observed bundles.

Rationalization by the Constructed Utility

For each observation i , suppose x is affordable at prices p^i :

$$p^i x \leq p^i x^i.$$

Then the i th affine term gives

$$u(x) \leq u^i + \lambda^i p^i (x - x^i) \leq u^i = u(x^i).$$

Thus x^i maximizes u on its budget set. The constructed utility rationalizes the observed demand choices.

Where the Afriat Numbers Come From

Let

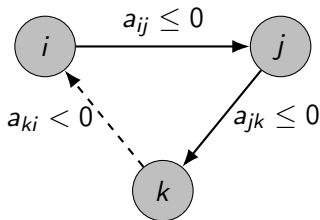
$$a_{ij} = p^i(x^j - x^i).$$

Then:

- $a_{ij} \leq 0$ means x^j was affordable when x^i was chosen.
- $a_{ij} < 0$ means it was strictly cheaper.

GARP says: if a directed cycle uses affordable comparisons, all links in the cycle must be weak links with zero slack. The proof of $(2) \Rightarrow (3)$ turns this no-strict-cycle condition into numbers u^i and λ^i satisfying Afriat's inequalities.

A Graph View of GARP



A cycle with one strict affordability link violates GARP. This is the budget-data analogue of a strict revealed-preference cycle.

Testing Axioms in Practice

- Representation theorems give exact empirical content:

D satisfies the axiom $\iff D$ is rationalizable.

- But exact tests are usually too coarse in applications.
 - One small mistake rejects the whole model.
 - A weak design may make violations almost impossible anyway.
- Two distinct questions:

Fit: How close is D to rationalizability?

Power: How hard was the test to pass?

Exact Rationalizability Is Binary

- Standard revealed preference gives:

$$D \in \mathcal{R} \quad \text{or} \quad D \notin \mathcal{R},$$

where $\mathcal{R}$ is the set of datasets rationalizable by utility maximization.

- This is useful for theory, but crude for measurement.
- If two subjects both violate GARP:
 - one may violate it once and by a tiny amount;
 - another may violate it repeatedly and severely.
- A binary pass/fail test treats them the same.

See Afriat (1967), Varian (1990), Houtman and Maks (1985).

Two Violators of Rationality

Person A

$$\begin{aligned}C_A(\{x, y\}) &= \{x\} \\C_A(\{x, y, z\}) &= \{z\} \\C_A(\{x, z\}) &= \{z\} \\C_A(\{y, z\}) &= \{y\} \\C_A(\{x, y, w\}) &= \{w\}\end{aligned}$$

Person B

$$\begin{aligned}C_B(\{x, y\}) &= \{x\} \\C_B(\{x, y, z\}) &= \{z\} \\C_B(\{x, z\}) &= \{z\} \\C_B(\{y, z\}) &= \{y\} \\C_B(\{x, y, w\}) &= \{y\}\end{aligned}$$

$$HM_A = \frac{4}{5}$$

$$HM_B = \frac{3}{5}$$

- Both violate exact rationality.
- Person A is closer to rationality in the Houtman–Maks sense.
- Dean–Martin asks a sharper monetary version:

what is the least costly set of RP implications to delete?

The Basic Fit Problem

- A goodness-of-fit measure asks for a distance:

$$\text{dist}(D, \mathcal{R}).$$

- Different indices define “distance” differently.

Houtman–Maks	: delete observations
Afriat/CCEI	: relax budgets
Money Pump	: measure exploitability of cycles
Dean–Martin	: delete costly RP implications

- There is no unique theory-free scalar called “degree of irrationality.”
- Each index answers a different repair question.

Back to Afriat: The Graph Structure

- Observations:

$$D = \{(p^i, x^i)\}_{i=1}^N.$$

- Define the expenditure slack:

$$a_{ij} = p^j(x^j - x^i).$$

- If

$$a_{ij} \leq 0,$$

then x^j was affordable when x^i was chosen.

- Hence:

$$x^i R^D x^j.$$

- The data generate a directed revealed-preference graph.

Afriat's Inequalities

- Afriat rationalizability requires numbers

$$u^1, \dots, u^N \quad \text{and} \quad \lambda^1, \dots, \lambda^N > 0$$

such that, for all i, j ,

$$u^j \leq u^i + \lambda^i p^i (x^j - x^i).$$

- Equivalently:

$$u^j - u^i \leq \lambda^i a_{ij}.$$

- This is a system of difference constraints.
- GARP rules out the cycle structures that make this system infeasible.

What a Violation Means in the Graph

- If x^i is revealed preferred to x^j , then rationality requires:

$$u^i \geq u^j.$$

- A strict revealed-preference cycle requires:

$$u^1 > u^2 > \dots > u^K > u^1,$$

which is impossible.

- So a revealed-preference violation is an inconsistency in the graph-implied utility ordering.
- Dean–Martin start exactly here:
if the graph is inconsistent, what is the cheapest way to repair it?

Dean–Martin: Edge Costs

- For every directly revealed-preference edge $i \rightarrow j$, define:

$$c_{ij} = p^j(x^i - x^j) = -a_{ij}.$$

- Since x^j was affordable when x^i was chosen,

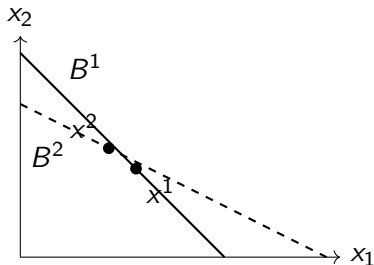
$$c_{ij} \geq 0.$$

- Interpretation:

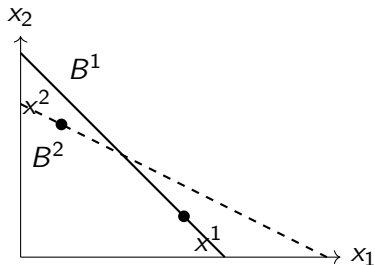
$$c_{ij} = \text{money left on the table by choosing } x^i \text{ instead of } x^j.$$

- Larger c_{ij} means the revealed-preference implication is more costly to discard.

Same Logical Violation, Different Magnitude



small violation



large violation

$$x^1 R^D x^2, \quad x^2 P^D x^1$$

$$x^1 R^D x^2, \quad x^2 P^D x^1$$

- Exact GARP rejects both datasets.
- Dean–Martin keeps track of how costly the offending revealed-preference edges are.

The Minimum Cost Index

- Let R^D be the set of directly revealed-preference edges.
- Choose a set $B \subseteq R^D$ of edges to delete.
- The cost of deleting B is:

$$\sum_{(i,j) \in B} c_{ij} = \sum_{(i,j) \in B} p^i(x^i - x^j).$$

- Dean–Martin choose the cheapest deletion set that destroys all cycles:

$$\min_{B \subseteq R^D} \left\{ \sum_{(i,j) \in B} c_{ij} : R^D \setminus B \text{ is acyclic} \right\}.$$

Normalized Minimum Cost Index

- The index normalizes the minimum repair cost by total expenditure:

$$MCI(D) = \frac{\min_{B \subseteq R^D} \left\{ \sum_{(i,j) \in B} p^i (x^i - x^j) : R^D \setminus B \text{ is acyclic} \right\}}{\sum_{i=1}^N p^i x^i}.$$

- Interpretation:

$$MCI(D) = 0 \iff D \text{ is rationalizable.}$$

- Higher values mean a larger monetary repair is needed.
- This captures both:
 - how many RP implications must be removed;
 - how severe those implications are.

Connection to the Afriat Proof

- Afriat asks whether all inequalities can hold:

$$u^j - u^i \leq \lambda^i a_{ij}.$$

- Dean–Martin ask which revealed-preference implications to drop so that the remaining graph can again be ordered.
- Same primitive object:

$$a_{ij} = p^i(x^j - x^i).$$

- Afriat uses the sign of a_{ij} to define consistency.
- Dean–Martin use the magnitude:

$$-a_{ij} = p^i(x^i - x^j)$$

as the cost of ignoring that revealed-preference edge.

A Simple Graph View

- Think of the data as a weighted directed graph:

$$i \rightarrow j \quad \text{if} \quad p^i x^j \leq p^j x^i.$$

- Edge weight:

$$c_{ij} = p^j(x^i - x^j).$$

- Exact rationality:

no forbidden cycles.

- Dean–Martin:

minimum total edge weight whose removal kills all cycles.

- This is a weighted cycle-hitting problem.

Why This Differs from Afriat

- The Afriat Index relaxes affordability comparisons by contracting budgets:

$$p^i x^j \leq e p^i x^i.$$

- It asks:
how much must budgets be relaxed to remove the worst violation?
- Dean–Martin instead ask:
which revealed-preference implications should be discarded at minimum monetary cost?
- CCEI is sensitive to the worst binding violation.
- MCI is sensitive to the aggregate cost of repairing all cycles.

Comparison of Fit Measures

Measure	Repair operation	What it captures
Houtman–Maks	delete observations	frequency of inconsistency
CCEI	relax budgets	worst violation magnitude
Money Pump	price cycles	exploitability of cycles
Dean–Martin	delete RP edges	number and cost of violations

- These measures are complementary, not interchangeable.
- They disagree because they define “closeness to rationality” differently.

Power: Fit Is Not Enough

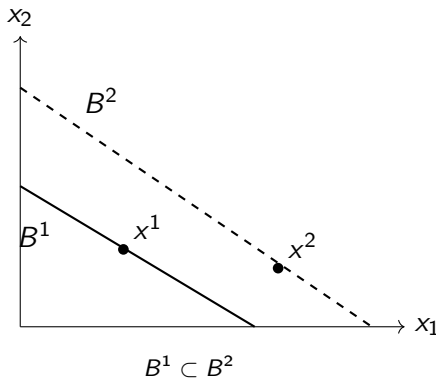
- A dataset may pass GARP because choices are rational.
- Or it may pass because the budget sets barely create opportunities for violations.
- Hence we also need to ask:

How restrictive was the test?

- Bronars' idea:
 - hold the budget sets fixed;
 - generate random choices from those same budgets;
 - compare the pass rate of actual data to the pass rate of random data.
- If random choice often passes, the revealed-preference test has low power.

Bronars (1987); Beatty and Crawford (2011).

Low Power: Nested Budgets



- $x^1 \in B^1$, so $x^1 \in B^2$.
- Hence choosing x^2 from B^2 can reveal:

$$x^2 R^D x^1.$$

- But $x^2 \notin B^1$, so choosing x^1 from B^1 does not reveal:

$$x^1 R^D x^2.$$

- No two-way revealed-preference cycle is possible.

Nested budgets have low power: even arbitrary choices may be unable to violate GARP.

Why This Matters for Behavioral Economics

- Behavioral models usually expand the rationalizable set:

$$\mathcal{R}_{\text{standard}} \subseteq \mathcal{R}_{\text{behavioral}}.$$

- That makes fit easier.
- So the relevant question is not only:

$$D \in \mathcal{R}_{\text{behavioral}}?$$

- It is also:

How much better is the fit, and how much flexibility bought it?

- Fit measures ask how close the data are to a model.
- Power/restrictiveness measures ask how much the model rules out.